Improving Age Diversity in Alzheimer's Clinical Drug Trials

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14 million

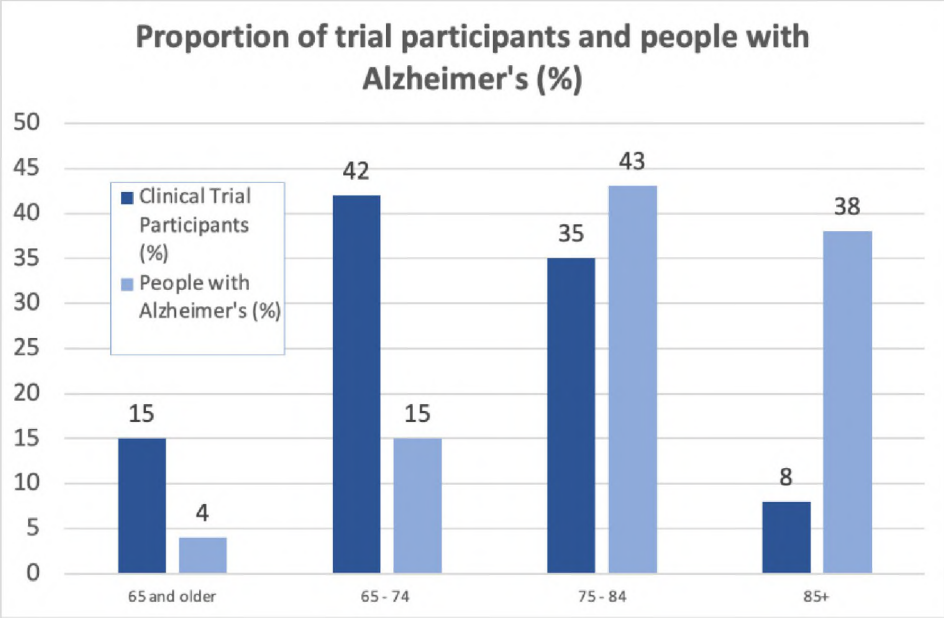
Projected Alzheimer's patients who are 65+ by 2060

96%

of the people affected by Alzheimer's are aged 65+

85%

of all clinical trial participants fall within the same age cohort



Consequences of under-representation

- Misunderstanding of age-related Pharmacokinetics & Pharmacodynamics
- Increased risk of adverse drug reactions
- Reputational consequences for sponsors

Drivers behind our Research

- FDA's mandate, current guidelines and regulations
- Aging population
- Downstream impact on developing world

GOAL

The FDA has stated an expectation that *"sponsors enroll participants who reflect the demographic for clinically relevant populations with regard to age, gender, race and ethnicity".* - Collection of Race and Ethnicity Data in Clinical Trials guidance by FDA in 2016

Participation to Prevalence Ratio (PPR) :

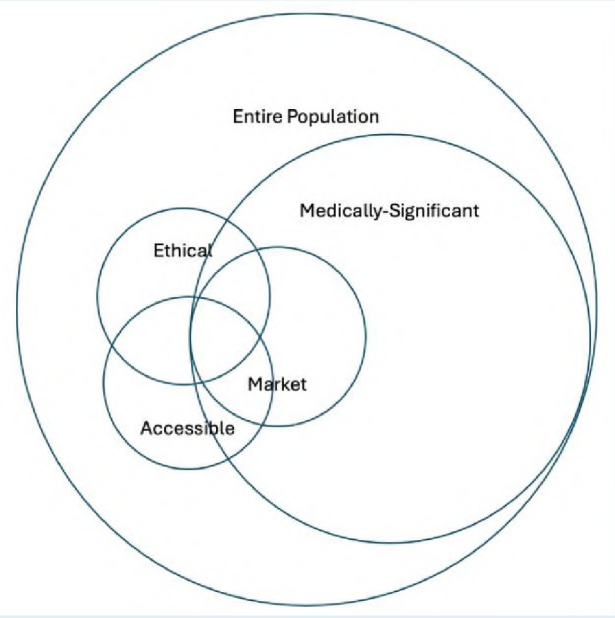
$$= \frac{\text{Participants belonging to the Specific Age Category / All participants in the trial}}{\text{Percentage of patients by age group among the } \textbf{medically significant and ethical population}}$$

The PPR for each disease should be approximately 1.

96% of the people affected by Alzheimer's are aged 65 and above, and only **85%** of all clinical trial participants fall within the same age cohort. Consequently, the computed PPR for the age group of 65 and above stands at **0.88.**

Potential Benchmarks

1. Country-Level Demographics
2. State or county level demographics
3. Market population (the people who are likely to have the economic capacity or insurance to eventually get the medical treatment)
4. The accessible population (people who, realistically, can enroll in a study with reasonable accommodations)
5. **The medically significant population**
6. **The ethical population (people whose participation would not raise questions of exploitation or undue risk).**

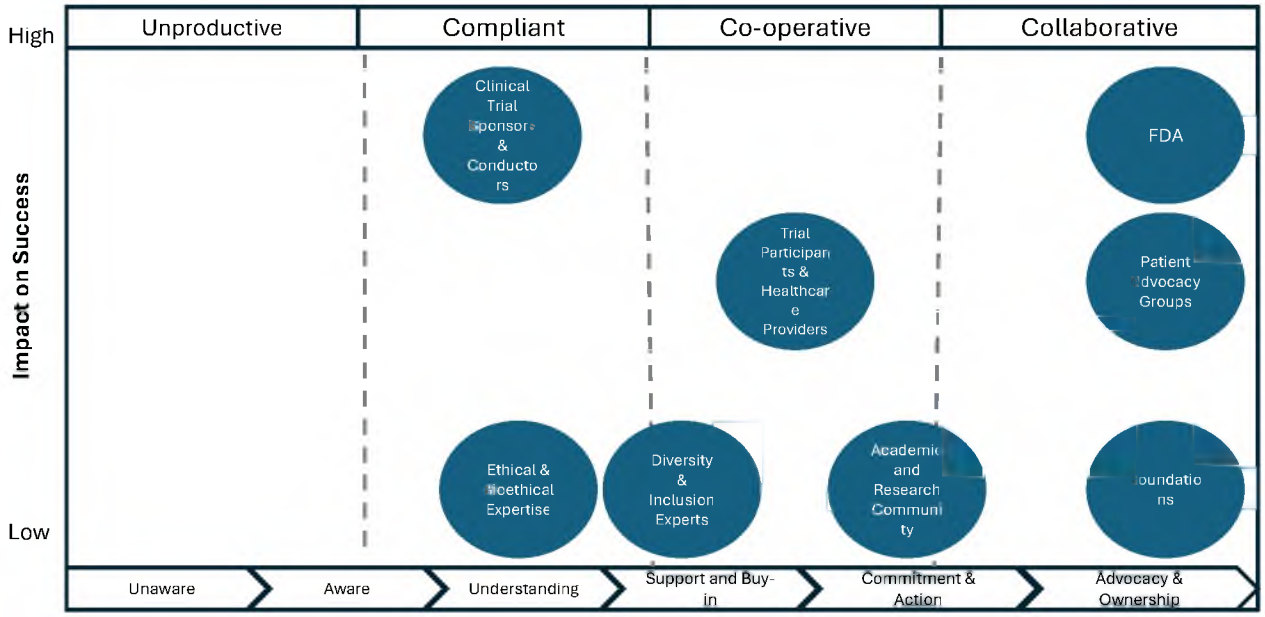


The medically-significant population sits inside the entire population and encompasses the market population. Accessible and ethical populations can be medically significant (but not necessarily) since trial phases are conducted across healthy individuals and patients. We should consider the intersection between medically-significant and ethical populations as a benchmark.

FINDINGS

We conducted interviews with key stakeholders to test our hypothesis drawn from our research and found the following:

	Hypothesis	Finding
1	**Rigid Clinical Trial Design Limits Age Diversity**	• Comorbidities and rigid inclusion/exclusion criteria are significant barriers to age diversity • Clinical trial phases must maintain scientific rigor without hindering age diversity. Adjustments, not relaxations, are needed
2	**Regulatory and Policy Constraints Affect Age Diversity in Trials**	• FDA guidelines are key for trial diversity; policy autonomy could drive design variety with sponsor commitment. • Congressional action could shape regulations, yet incentivizing lawmakers remains challenging.
3	**Incorporating seniors into trials faces operational and economic hurdles**	• Diverse age group inclusion in trials faces financial and logistical issues • Strategic planning and reimbursements can address issues, but may slow drug development, reducing sponsor focus
4	**Cultural Perceptions Influence Participant Recruitment/Retention**	• Research has shown that societal and cultural barriers do exist, but targeted strategies such as relationship building and trust cultivation may help overcome them
5	**Improving elderly inclusion in clinical trials demands collaboration across stakeholders**	• A holistic approach integrating key stakeholders is essential, from enrollment through trial completion • Challenges in trial design and the need for regulatory changes imply that external interventions might be more impactful than internal holistic approaches



Driving change in clinical trial age diversity will be driven primarily by the FDA and Patient Advocacy Groups

RECOMMENDATIONS

Our recommendations will drive the participation to prevalence ratio closer to one, improving the efficacy and safety of Alzheimer's treatments for older adults

1) Provide greater incentives for sponsors

- Technically Correct
- Administratively Feasible
- Politically Supportable

2) Standardize Informed Consent Processes for patients

3) Mandate age-specific Pharmacokinetic disclosures

4) Implement age-friendly communication standards

5) Enhance FDA monitoring of age representation in trials

RISK MITIGATION

Recommendation	Risk Mitigation Strategy
1	Increase administrative feasibility by gradually introducing the program in stages to allow the FDA and trial sponsors time to adapt
2	Increase administrative feasibility by collaborating closely with key stakeholders during the guideline development process. Increase political supportability by engaging with patient advocacy groups and Alzheimer's communities
3	Increase administrative feasibility by providing clear, step-by-step guidance for pharmaceutical companies. Increase political supportability by conducting outreach to explain the public health benefits
5	Increase administrative feasibility by developing efficient reporting tools and provide training for sponsors. Increase political supportability by highlighting the importance of this data through engaging advocacy

International Application

The FDA has an opportunity to help clinical trials regulators in the developing world develop context-specific guidelines to improve age diversity in Alzheimer's clinical trials globally

- Strengthen Research Capacity and Infrastructure
- Foster Collaborative Networks
- Engage Community and Encourage Professional Development